

When you create a button with `tkinter`, you must pass in a function (or a method of an object), which will be called back when the user clicks the button. Here, we are passing `myFunction` as the function to be called back. (This call is using keyword parameters, which will be discussed at length in Chapter 7.) The `tkinter` button remembers that function as the callback, and when the user clicks the resulting button, it calls the function `myFunction()`.

You can also use a callback when you initiate some action that may take some time. Instead of waiting for the action to finish and causing the program appear to freeze for a period of time, you provide a callback to be called when the action is completed. For example, imagine that you want to make a request across the internet. Rather than making a call and waiting for that call to return data, which may take a long time, there are packages that allow you to use the approach of making the call and setting a callback. That way, the program can continue running, and the user is not locked out of it. This often involves multiple Python threads and is beyond the scope of this book, but the technique of using a callback is the general way that it is done.

Using a Callback with SimpleButton

To demonstrate this concept, we'll make a minor modification to the `SimpleButton` class to allow it to accept a callback. As an additional optional parameter, the caller can provide a function or method of an object to be called back when a click on a `SimpleButton` object happens. Each instance of `SimpleButton` remembers the callback in an instance variable. When the user completes a click, the instance of `SimpleButton` calls the callback.

The main program in [Listing 6-8](#) creates three instances of the `SimpleButton` class, each of which handles the button click in a different way. The first button, `oButtonA`, provides no callback; `oButtonB` provides a callback to a function; and `oButtonC` specifies a callback to a method of an object.

File:

**PygameDemo9_SimpleButtonWithCallback/Main_SimpleButton
Callback.py**

```
# pygame demo 9 - 3-button test with callbacks
```